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MILITARY STANDARD

WATER PURIFICATION COMPOUNDS

FSC 6850

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DEPARTMENT OF DEFENSE
Washington, D. C. 20301

Water Purification Compounds

MIL-STD-1215A

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to Commander, Edgewood Arsenal, Attn: SMUEA-DE-ES, Aberdeen Proving Ground, Maryland 21010.

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FOREWORD

This is the first book format standard generated on water purification compounds. This standard is mandatory for use by all departments and agencies of the Department of Defense in the selection of items for application. It is intended to prevent the entry of unnecessary items (sizes, types, varieties) into the Department of Defense logistics system. This is not a procurement document.

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1. SCOPE

1.1 Coverage. This standard is a comprehensive presentation of nomenclature, symbols, physical and chemical properties, military use, packaging data, labeling, general safety precautions, storage information, and shelf life of military water purification compounds. This standard does not necessarily include all classification of the items represented by the title or those which are commercially available. It does contain items preferred for use in the selection of water purification compounds for application by the Department of Defense. This standard covers the following three items:

Name	No. of Items
CHLORINATION KIT, WATER PURIFICATION	2
WATER PURIFICATION TABLETS, IODINE	1

1.2 Application. Items listed herein accommodate essential requirements of the military and defense agencies, and will effect continued economies in all logistics functions when properly employed in new applications.

2. REFERENCED DOCUMENTS

The issue of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

Federal Specifications

O-C-114	Calcium Hypochlorite, Technical and Chlorinated Lime, Technical
O-C- 289	Chlorination Kit, Water Purification, (for Treatment of Drinking Water)
PPP-C-301	Chemicals, Dry, and Paste, Packaging and Packing of

Military Specifications

MIL-W-283	Water Purification Tablet, Iodine

Rules and Regulations

ICC Shipping Regulations

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3. GLOSSARY

3.1 Definitions.

Anhydrous - Pertaining to a salt which has no water of hydration present in the crystaline structure.

Hermetically - Made air-tight or perfectly closed.

Solubility - The weight of a substance which will dissolve in a specific volume of solvent at a specific temperature to produce a saturated solution. In this standard, the solubility will be expressed as the weight of solvent in grams dissolved in 100 ml of the specific solvent.

3.2 Abbreviations. The same abbreviation is used for all tenses, the possessive case, and the singular and plural forms of a given word.

g - gram

ICC - Interstate Commerce Commission

mg - Milligram

USP - United States Pharmacopeia

4. GENERAL REQUIREMENTS

4.1 Packaging data and labeling. All dry and paste chemicals included in this standard shall be packaged in accordance with PPP-C-301, and all applicable documents mentioned in this standard.

4.2 Percent by weight. All percentages noted are understood to be maximum percent by weight unless otherwise indicated.

4.3 Shelf life. Factors such as moisture, temperature, type and condition of container, exposure to sunlight and the atmosphere cause variations in shelf life. Ideal storage conditions are outlined for each item. An approximate period of time after which this material will no longer be suitable for its intended use is also presented. The term "cool" denotes temperatures from above freezing up to 110 degrees Fahrenheit, when stored out of direct sunlight. The term dry is usually used to denote an area where condensation does not come in contact with the packages or contents (for example, storing on pallets away from walls in an enclosure or building). Periodic examinations of the container or material should be made more frequently when storage conditions vary from the ideal. For applications where quality may be critical each compound should be analyzed prior to use. Shelf life is dated from the date of manufacture. All chemicals in this standard shall not be older than one year from the date of manufacture when purchased except where specified otherwise under storage data.

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4.4 Safety. General laboratory safety and hygienic measures should be exercised in the use of these chemicals. Only general safety precautions are outlined in this standard. For more specific information the appropriate safety or medical authorities should be consulted for the proper personal protective measures and environmental controls.

5. DETAIL REQUIREMENTS

5.1 Name. WATER PURIFICATION TABLETS, IODINE

5.1.1 Specifications. Military Specification MIL-W-283, Water Purification Tablets, Iodine.

5.1.2 Technical description. Iodine water purification tablets are of a double convex type of the following ingredients: tetraglycine hydroperiodide, sodium acid pyrophosphate anhydrous, and talc. The composition and requirements of the iodine tablets are specified in the following three tables:

TABLE I. - Composition of tablets by weight

Ingredient	Minimum Milligrams	Maximum Milligrams
Sodium acid pyrophosphate, anhydrous	82.5	92.3
Talc, USP	----	6.0
Tetraglycine hydroperiodide	19.3	21.3

TABLE II. - Requirements for tetraglycine hydroperiodide

Ingredient	Minimum Percent	Maximum Percent
Acitive iodine	39.5	42.6
Ash	----	0.5
Titratable iodine after 7 days storage in a tightly sealed bottle at 130°F	34.5	----
Total nitrogen	7.35	7.85

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TABLE III. - Requirements for sodium acid pyrophosphate, anhydrous

Ingredient	Minimum Percent	Maxiumm Parts Per Million
Sodium acid pyrophosphate, anydrous	95.0	----
Arsenic	----	0.5
Fluorine	----	20
Iron phosphate plus aluminum phosphate	----	300
Lead	----	2.0
Moisture	----	1000.0

The maximum weight limitation for the individual tablet is 120 milligrams and the maximum moisture content is 0.2 percent. When tested for titratable iodine, each tablet shall average between 7.60 and 9.00 milligrams. Individual tablet weight deviations must conform to the tolerances for tablet weights as outlined in the Weight Variation Test for Tablets referenced in the Pharmacopeia of the United States of America.

5.1.3 Use data. Iodine water purification tablets are intended for military use for the purification of water in canteens. This item is issued for emergency health protection reasons to provide troops in the field with drinking water free from disease bearing organisms.

5.1.4 Packaging data and labeling. For military use iodine water purification tablets are packaged in unit quantity bottles containing 50 tablets. The space between the tablets and bottle closure is filled with clean fiberglass. Each bottle must have a moisture vapor resistant closure. The bottle can be either round or square made of nonactinic glass. For intermediate packaging, one hundred bottles containing 50 tablets are packed in a snug-fitting fiberboard box. For overseas shipment, 12 intermediate boxes containing 100 bottles of 50 tablets each are packaged in snug-fitting fiberboard containers. For domestic immediate use packaging, the tablets are packaged according to the normal practice of the manufacturer. Containers are to be in accordance with the rules and regulations of carriers applicable to the means of transportation. The intermediate packages and shipping containers shall be marked to show the lot number, the name of the supplier, and the contract number.

5.1.5 Safety precautions. There are no particular safety precautions to follow regarding iodine water purification tablets.

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5.1.6 Storage data. Not more than 6 months shall have elapsed from date of manufacture of the tablets, to date of delivery. The tablets must be maintained in a dry condition. The tetraglycine hydroperiodide component should not be stored at temperatures above 90 degrees Fahrenheit. The shelf life of the tablets is 36 months but will be substantially reduced by moisture, high humidity, and high temperature. When the issue date of this material is greater than two years after the date of manufacture, samples of the iodine tablets will be assayed for titratable iodine and solubility.

5.2 Name. CHLORINATION KIT, WATER PURIFICATION

5.2.1 Specifications. Federal Specification O-C-289, Chlorination Kits, Water Purification, (For Treatment of Drinking Water).

5.2.2 Technical description. The two chlorination kits consist of components indicated in Table IV.

TABLE IV. - Chlorination kit components

	Components	Quantity
Without Sodium Sulfite for dechlorination	Tubes of calcium hypochlorite	100
	Test kit, 1 mg/l of total chlorine	1
	Test kit, 5 mg/l of total chlorine	1
	Test kit, 10mg/l of total chlorine	1
With Sodium Sulfite for dechlorination	Tubes of calcium hypochlorite	100
	Test kit, 6 mg/l of total chlorine	2
	Sodium sulfite tablets	2

The tubes of calcium hypochlorite must conform to the requirements of O-C-114 for properties listed therein. The minimum available chlorine is 65 percent by weight. The maximum moisture content shall be 10 percent by weight and the available chlorine loss when heated at 100° ± 2°C for two hours is not to exceed 10 percent by weight. The glass tubes are to be clear with an outside diameter of 8.55 mm to 8.95 mm and a wall thickness of 0.8 ± 0.05 mm. The tubes are hermetically sealed by fusion of the glass to prevent leaks. The tubes must have a transverse file mark near the center to allow a clean right angle break. Each tube contains 0.5 ± 0.1 grams of calcium hypochlorite. The orthotolidien dihydrochloride tablet test kit consists of a clear plastic vial with threaded top and 50 orthotolidine dihydrochloride tablets in a screw-top flint glass or nonactinic bottle. The weight of the sodium sulphite tablet is 2.6 ± 0.2 grams.

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5.2.3 Use data. The water purification chlorination kit without sodium sulfite for dechlorination is intended for military use in the disinfection of water in lister bags and small containers by chlorine in order to render water safe for drinking purposes. Kit with sodium sulfite for dechlorination is for use by US Navy.

5.2.4 Packaging data and labeling. For military use procurement documents should specify the type of packaging required, and the applicable level of protection based on the ultimate destination and storage conditions. One hundred tubes of calcium hypochlorite, three testing kits and an instruction sheet shall be packaged as a unit. One hundred tubes of calcium hypochlorite, two testing kits, one hundred sodium sulfite tablets, and an instruction sheet shall be packaged as a unit. Set up paperboard boxes of the complete units shall be marked or labeled with the name and quantity of the chemical contained therein, kit stock number, contract number, and the name of the supplier. The box shall also be marked in minimum one quarter-inch high letters as follows:

Chlorination kit (water purification) Type_____

5.2.5 Safety precautions. There are no particular safety precautions in connection with the handling of the water purification kits.

5.2.6 Storage data. The shelf life of the water purification kit is limited to two years based on the shelf life of calcium hypochlorite.

Notice. - Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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